CBIO Hackathon Report: Leveraging Plasma cfDNA H3K4me3 ChIP-seq to Differentiate Healthy from Liver-Diseased Individuals and Classify Disease Types

Hillel Darvish
darvish.hillel@mail.huji.ac.il

Amit Alper
amit.alper@mail.huji.ac.il

Ilay Gussarsky
ilay.gussarsky@mail.huji.ac.il

David Gershelis
david.gershelis@mail.huji.ac.il

Ghasan Jaber
ghasan.jaber@mail.huji.ac.il

1 Introduction

In recent years, Cell-Free DNA (cfDNA) has revolutionized the field of noninvasive diagnostics in various fields, such as oncology and organ transplantation, and shows promising progress in liver diseases (Sorbini et al., 2024; Barrera-Saldaña et al., 2021).

Cell-free DNA refers to DNA fragments that were released into the blood stream, originating from cells that undergo cell death (Corcoran & Chabner, 2018).

Within the nucleus, DNA is spatially organized around protein complexes called nucleosomes. Each nucleosome consists of subunits known as histones, which undergo post-translational modifications, collectively referred to as the histone code (Jenuwein & Allis, 2001). These histone modifications are associated with various biological functions. For instance, some mark enhancer regions, others mark promoter regions of silenced genes, or promoter regions of genes that are expressed, along with other biological roles (Jenuwein & Allis, 2001). One modification of interest is H3K4me3, which is linked to promoter regions of genes that can potentially be expressed (Bannister & Kouzarides, 2011). When cells die and release their DNA into the bloodstream, DNA associated with nucleosomes is also released, including those with histones marked by this modification.

By analyzing this DNA, using ChIP-seq (Sadeh et al., 2021), each sequencing read can be mapped to a specific gene. The collection of all reads within a sample creates a profile of the

genes expressed in the cells that died and released their DNA. Therefore, we hope to use this gene expression profile to derive interesting conclusions. Among these genes, some may serve as markers for the presence of disease, specifically markers of liver disease.

In this hackathon project, we aim to use data of H3K4me3 ChIP-seq on plasma cell free DNA to answer on two main objectives: Can we differentiate between healthy and sick liver patients? And can we differentiate between different liver disease pathologies?

We will develop a multiple classifiers capable of distinguishing between healthy and sick individuals. We will test different feature selection methods, to sort genes by their ability to differentiate between the groups. Then we will use the selected genes as features for these classifiers. Furthermore, we will extend our approach to distinguish between different liver disease pathologies, focusing on Metabolic Dysfunction-Associated Steatohepatitis (MASH), Primary Biliary Cholangitis (PBC) and Autoimmune Hepatitis (AIH).

2 Data

The data was collected by Prof. Nir Friedman's laboratory. Blood sample donations were collected, from which, plasma was isolated. The plasma went through ChIP-seq experiment to capture histones using specific antibodies for H3K4me3 modification. DNA associated with the isolated histones was sequenced. The resulting reads were aligned to the humam genome, and each read was assigned to the gene it best matched. The dataset comprises three tables: **Healthy Individuals Table:** Each column corresponds to a ChIP-seq experiment for the H3K4me3 histone signature in a particular individual, and each row indicates the normalized read count for a specific gene. This table contains 715 samples (columns) and 18,182 genes (rows). **Patients Table:** This table includes the same gene rows as the healthy individuals table but consists of 207 patient samples. **Clinical Data Table:** This table provides clinical information for 188 of the 207 patients, including biopsy result details.

3 Methods

3.1 Feature Selection

A sub-goal in our project was to find genes whose read counts can help us differentiate between sick and healthy individuals. We will call those genes *feature genes*. We tried four different methods for feature selection: t-test, Kolmogorov-Smirnov, distances between Gaussians approach, and a neural network based approach. In the first three approaches, the main principle is to find genes that the distance in their expression levels between the two groups is the highest.

3.1.1 t-test approach

The student's t-test is a statistical test to determine whether the difference between two result groups is statistically significant or not. We used t-test for independence, which compares the means of two groups, and states whether there is a significant difference between the means. For each gene, we conducted the t-test for independence and computed the p-value of the test, where the groups were the healthy group and the sick group. The selected features are the n genes with the lowest p-value. t-test for independence assumes that the data originates from a normal distribution, which might not be the case in our data. However, t-test is fairly robust to violation of those assumptions (Stonehouse and Forrester, 1998).

3.1.2 Kolmogorov Smirnov test

The Kolmogorov-Smirnov test (KS-test) tests the difference between two distributions. Given two cumulative distribution functions (CDFs) G, F, the KS-statistic D is defined as the largest vertical distance between the two CDFs: $D = \sup_x |F(x) - G(x)|$. The KS-test checks whether the distributions of a gene's expression in healthy vs. sick individuals significantly differ. As in the previous feature selection method, for each gene we will compute the p-value of the test and select the n genes as the genes with the lowest p values. The lower the p-value, the larger the difference between the distributions, which can lead to better differentiation between healthy and sick samples.

3.1.3 Gaussian Overlap Approach

This approach is closely related to the mean classifier described below. The underlying assumption is that for two distributions to be differentiable, their overlap has to be minimal. It is trivial that given one feature, the overlap between the two hypothesis in that feature's value's distribution is some kind of blind spot where it is difficult to distinguish between them. However, generally, small overlap doesn't mean it is trivial to distinguish between the two hypothesis: Figure 1 shows an example for two Gaussian pairs with similar overlap, but where it is simpler to find a classifier in the right example. More specifically, a threshold-based classifier will do much better on the example on the right. The PDF overlap approach, for each feature, es-

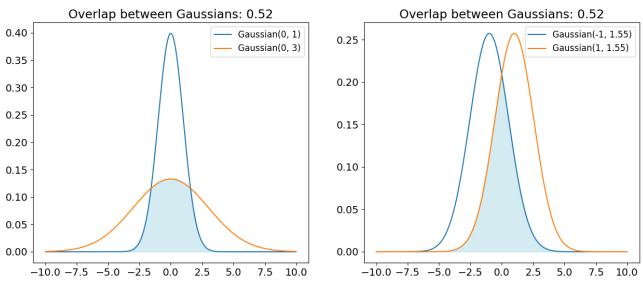


Figure 1: Gaussian pairs with similar overlap, but different overall behavior

timates its sick/healthy distributions with a Gaussian distribution and gives a score for the feature based on the (negative) overlap between those two Gaussians.

3.1.4 Neural Network based approach

Neural network architectures vary widely, each designed for specific types of tasks. In our case, each data point consists of a binary label ("sick" or "healthy") and a vector of numerical features, all of which have the same units and are normalized. Given this structure, we decided to try and score the genes based on a calculation, using the learned weights and performance of a simple Fully-Conected Feed-Forward Neural-Net (shorten here as: FNN) with one hidden layer. This architecture is ideal for datasets where the relationships between features are unknown, and the output is a single value, which represents the probability of being sick.

Shown below is a visualization of the feature vector, which represents a single data point of $N_f = 18182$ genes each individual has. Next to it are the weights of the N_{nets} nets. Each net i has its own hidden-layer width, N^i_{width}, where each node j of those N^i_{width} has its weight vector W^i_j of size N_f. Each net also has an output node with its weights, one for each node j in the hidden layer. The ROCAUC from the test set for each net is also shown and used:

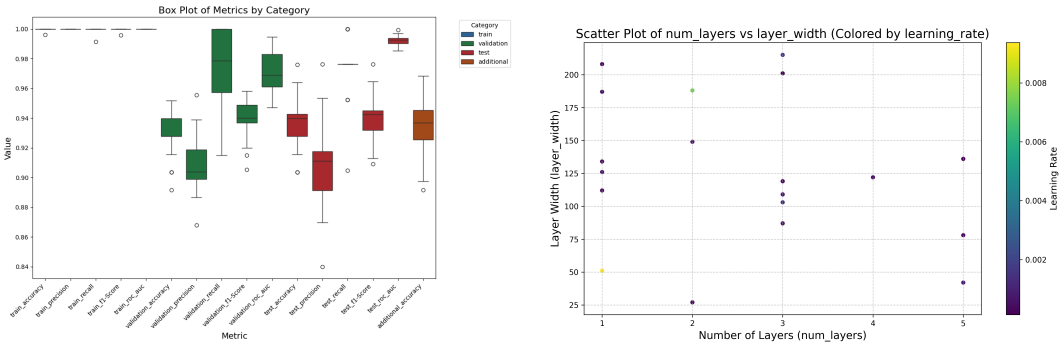
$$
\begin{array}{ccccc}
 & \text{Net 1} & & \text{Net Nnets} \\
 & \text{Hidden Nodes Weights} & & \text{Hidden Nodes Weights} \\
\text{Features Vector} & \begin{bmatrix} w^1_{1,1} \\ \vdots \\ w^1_{Nf,1} \end{bmatrix} \cdots \begin{bmatrix} w^1_{1,Nwidth^1} \\ \vdots \\ w^1_{Nf,Nwidth^1} \end{bmatrix} & \cdots & \begin{bmatrix} w^{Nnets}_{1,1} \\ \vdots \\ w^{Nnets}_{Nf,1} \end{bmatrix} \cdots \begin{bmatrix} w^{Nnets}_{1,Nwidth^{Nnets}} \\ \vdots \\ w^{Nnets}_{Nf,Nwidth^{Nnets}} \end{bmatrix} \\
\begin{bmatrix} f_1 \\ \vdots \\ f_{Nf} \end{bmatrix} & \text{Output Node Weights} & & \text{Output Node Weights} \\
 & \begin{bmatrix} o^1_1 & \cdots & o^1_{Nwidth^1} \end{bmatrix} & & \begin{bmatrix} o^{Nnets}_1 & \cdots & o^{Nnets}_{Nwidth^{Nnets}} \end{bmatrix} \\
 & \text{The ROCAUC}_1 \text{ score on the test set} & & \text{The ROCAUC}_{Nnets} \text{ score on the test set}
\end{array}
$$

The FNN scores is a vector of size Nf, which holds the scores of this feature selection approach. The idea behind it is using the weight learned for each gene, in each hidden node, in each hidden-layer from the one-hidden-layered nets. To account for the information each node in the hidden layer learned, and for its effect on the output node, and the overall performance of the net on the test set, a weighted-average of weighted-averages was calculated, as shown below:

$$
\text{ROCAUC}_{\text{sum}} = \sum_{i=1}^{Nnets} \text{ROCAUC}_i \text{ , for } i \in [1 \dots Nnets] : \quad \text{o}^{\text{i}}_{\text{sum}} = \sum_{j=1}^{\text{Nwidth}^i} \text{o}^i_j
$$

$$
\text{FNNscores} = \frac{1}{\text{ROCAUC}_{\text{sum}}} \sum_{i=1}^{\text{Nnets}} \text{ROCAUC}_i \left(\frac{1}{\text{o}^{\text{i}}_{\text{sum}}} \sum_{j=1}^{\text{Nwidth}^i} \text{o}^i_j \text{W}^i_j \right), \quad \text{where } W^i_j = \begin{bmatrix} w^i_{1,j} \\ \vdots \\ w^i_{Nf,j} \end{bmatrix}
$$

Weights were initialized randomly, and ReLU was used as the activation function on the output of each neuron. For the last layer we used the sigmoid function as the activation function, to output a probability. The loss function we chose is the difference between the numeric value of a data points label (0 or 1) to the output number from the net (a number between 0 and 1). We used the python library *optuna* that automatically tries running the net with different hyper-parameters from given ranges, and then optimizes for the best hyper-parameters. We ran this process 20 times, each time optuna ran the net 5 times, starting at random hyper-parameters from given ranges. From the 20 runs, 6 ended with FNNs with one hidden layer (Figure 2b). The performances of the 20 nets on the different data sets are similar as seen in Figure 2a, supporting the usage of nets with one hidden layer.



(a) Box plot of FNN results

(b) Scatter plot of network parameters

Figure 2: Results of the 20 FNNs. (a) The dataset was split into four parts: train, validation, test, and additional-test. Due to class imbalance, each sick sample was paired with a randomly selected healthy sample to form the train-validation-test groups, while the remaining healthy samples formed the additional-test set. The box plot shows score distributions across networks. (b) A scatter plot visualizing the number of hidden layers, their width, and the learning rate for each of the 20 networks.

3.2 Classification Between Healthy and Sick Samples

After finding lists of genes that differentiate sick and healthy samples better than others, we attempt to create different classifiers that use these features to distinguish between healthy and sick samples. In the following section, we will describe the different classification methods we have tried.

3.2.1 Random Forest

Our baseline model was a random forest classifier, without feature selection. The random forest classifier ensembles several decision trees, and this means there are many relevant hyper-parameters. Since this is a baseline model, we used a simple grid search, to optimize the hyper-parameters, as explained on Table 1.

Hyperparameter	Values
Number of estimators (*n_estimators*)	50, 100, 200
Maximum depth (*max_depth*)	Unlimited, 5, 10
Minimum samples split (*min_samples_split*)	2, 5

Table 1: Hyperparameter grid used for Random Forest. *Grid Search* was used to choose the best combination of hyper-parameters listed above, while using the default option for the other hyper-parameters.

3.2.2 K-Nearest Neighbors

A K-Nearest neighbors classifier was used as another simple classifier. however, for this classifier, feature selection methods were used. The number of neighbors was set to 5 and features were standardized using python's *StandardScaler*.

3.2.3 Mean Ensemble Classifier

The Mean Ensemble Classifier is a classifier that is based on the assumption that the selected feature genes have distinct mean expression levels between the healthy group and the sick group. The Input for the classifier is the gene expression levels in the healthy group and sick group, only for the selected genes. During training, the classifier computes the mean expression levels for each gene separately for healthy and sick samples:

1. Compute the mean expression levels in the healthy group and in the sick group, for each gene g in the selected feature set G (μ_H^g, μ_S^g)
2. Set a decision threshold at the midpoint between the two means: $T_g = \frac{\mu_H^g + \mu_S^g}{2}$

Prediction Rule: To classify a new sample x, the classifier evaluates each gene's expression level relative to its threshold T_g: $\hat{y_g} = \begin{cases} 1, & \text{if } \mu_H^g > \mu_S^g \text{ and } x_g < T_g \\ 1, & \text{if } \mu_H^g < \mu_S^g \text{ and } x_g \geq T_g \\ 0, & \text{otherwise} \end{cases}$

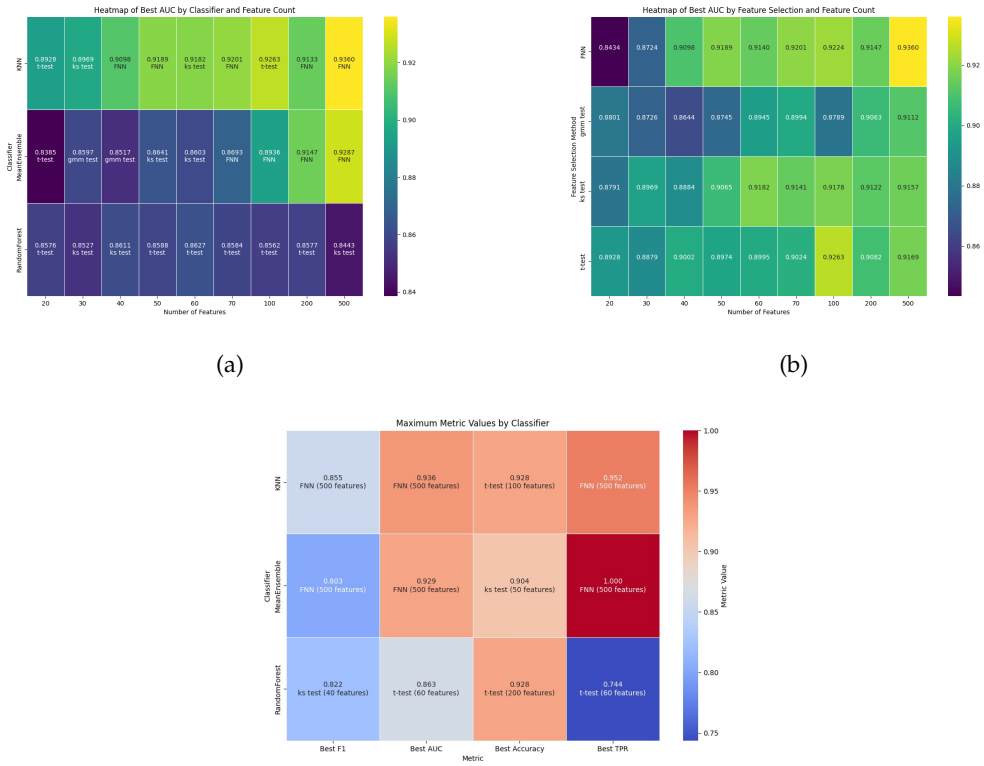
The prediction rule is the majority vote of the classifiers $\hat{y_g}$ for all genes g in the subset of selected feature genes G. As will be seen in the results section, the feature selection process is crucial for this classifier.

3.3 Cross validation

We have designed several feature selection methods. To find the best combination between method and selected genes for each classifier, we performed 5-fold cross-validation. AUC was the primary evaluation metric, alongside F1 score, True Positive Rate (TPR), and Accuracy.

4 Results

4.1 Cross Validation Results

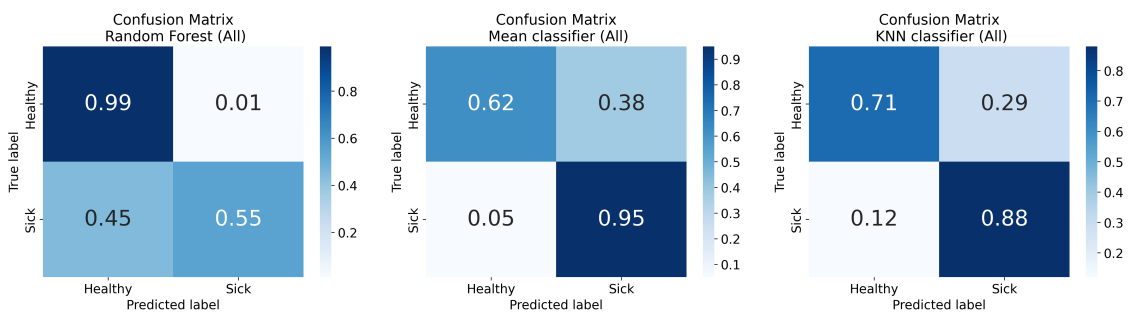


(a) (b)

(c)

Figure 3: Cross validation results. **(a)** Heatmap showing the best AUC for each classifier and feature count, along with the feature selection method that achieved it. **(b)** Heatmap showing the best AUC for each feature selection method and feature count. **(c)** Summary of cross-validation using different evaluation metrics (F1, AUC, Accuracy, and True Positive Rate). The best score for each classifier is shown, along with the feature selection method and number of features that achieved it.

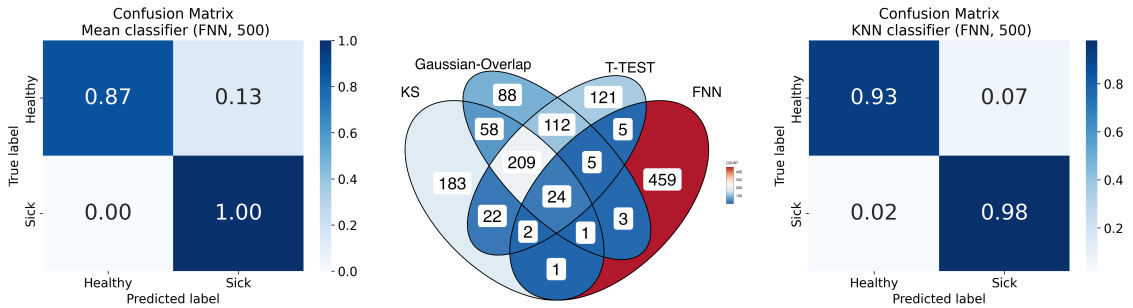
4.2 Baseline and Intermediate Results



(a) (b) (c)

Figure 4: Confusion matrices For the three classifiers, without feature selection methods.
(a) Random Forest classifier. **(b)** Mean Ensemble classifier. **(c)** KNN classifier.
Row-wise normalization was applied for all confusion matrices.

4.3 Final Results



(a) (b) (c)

Figure 5: **(a)** Confusion matrix for the Mean Ensemble classifier. Feature selection method was the FNN, with 500 genes. **(b)** Venn Diagram for the gene overlap between the four feature selection methods with 500 gene for each method. **(c)** Confusion matrix for the for the KNN classifier. Feature selection method was the FNN, with 500 genes.

5 Differentiating Between Different Liver Diseases

Given a classifier designed to distinguish between sick and healthy individuals, we aim to further classify between different liver diseases. The dataset, derived from patients who underwent liver biopsies, contains labels based on their respective diagnoses. The model we developed for this purpose operates under the assumption that all input samples are classified as "sick," meaning the task is to differentiate between individuals with liver diseases rather than distinguishing between sick and healthy individuals in the general population.

5.1 Feature Gene Selection

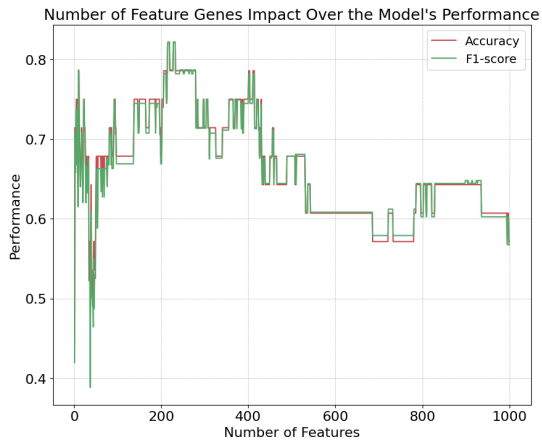
In this task we need to differentiate between new labels, therefore we need to look for new feature genes. As detailed in Section 3.1, there are multiple approaches for selecting feature genes that can differentiate between different liver diseases. For this problem, we have decided to use the Kolmogorov-Smirnov test (described in 3.1.2) to discover feature genes differentiating between the diseases.

The classifier structure also impacts the feature gene selection. We have decided to use the **one-vs-all** approach. The goal of this classifier is to classify the proper disease given a sick sample. Thus, we aimed to find differentiating genes between each disease and all the other diseases.

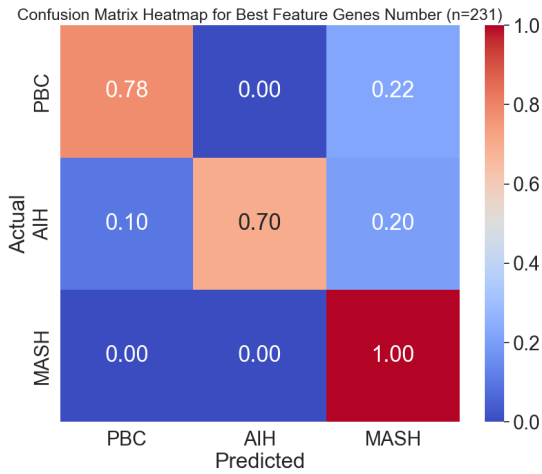
To validate our results, we applied a False Discovery Rate correction to the p-values. Surprisingly, after correction, the adjusted p-values did not show significant differences, indicating that the selected genes may not have as strong a discriminatory power as initially expected.

5.2 Classifier and Results

Balancing the Input Data Before describing our classifier, we will address an issue in our input data. The input data was imbalanced, meaning the number of samples per diagnosis varied significantly. To mitigate this, we decided to subsample the input data to ensure that all diagnosis classes had the same number of samples before splitting into train-test sets. An additional step was taken to ensure that all diagnosis classes were represented in both the training and testing sets.



(a)



(b)

Figure 6: Liver Disease Ensemble Classifier results. **(a)** Accuracy and F1 score of the Liver Disease Ensemble Classifier as a function of the number of selected feature genes. **(b)** Confusion matrix for the classifier that had the best accuracy (n=231). Row-wise normalization was applied.

The Classifier Using the feature genes identified earlier, we developed a classifier aimed at distinguishing between liver diseases. The classifier's decision rule is based on the formula: $\hat{y} = \max_i(P_i(x))$, where $P_i(x)$ is the probability predicted by the model of the i-the disease class, that a given feature vector x belongs to this specific disease class. In other words, we train simple *logistic regression* classifiers that are able to output whether a given sample comes from a certain disease or not. The ensemble then chooses the most probable disease based on the individual classifiers. It is important to note that each simple disease classifier is learning based on a different set of feature genes. As stated earlier, we calculate the best gene according to KS-test result for all possible one-vs-all comparisons. For each disease class classifier, we select the top n genes (those with the lowest p-values, as the adjusted p-values were less informative. All classifiers share the same n) and based on them a logistic regression classifier learns to classify between the i-th disease class to all the others. Each simple classifier is trained based on its specific set of genes. Classifiers were trained under the assumption that the samples comes from sick population.

6 Discussion

In this Hackathon project, we used H3K4me3 ChIP-seq on plasma cell-free DNA data, to distinguish between healthy individuals and liver disease patients, and further differentiate between different liver diseases.

To identify the most relevant genes for classification, we tested four feature selection methods and analyzed overlap among the top 500 ranked genes per each method. As shown in Figure 5b, only 24 genes were shared across all methods. The first three approaches (t-test, Kolmogorov-Smirnov, and Gaussian overlap) had an overlap of 209 genes, while the neural network method selected more unique genes than any other method (459 genes). This suggests that different methods capture distinct biological signals. Future research will focus on understanding these differences and evaluating the biological significance of the selected genes.

For classification, we tested three models (see methods section) and optimized both the feature selection method and the number of genes using cross-validation. The first analysis of cross validation was by using AUC as the evaluation score, and testing the best AUC results achieved for each classifier and number of genes. As shown in Figure 3a, strong performance was achieved even with fewer genes, indicating that a smaller, well-chosen subset can be effective. However, the best results were achieved by using the top 500 selected genes by the neural network. As expected, for a different number of genes, different feature selection methods were better. In addition, we wanted to evaluate the success of each feature selection method, with different number of features. As shown in Figure 3b, all feature selection methods showed similar results, with several exceptions. The neural network achieved the best overall results, but was less effective than the other methods on a small subset of genes. This analysis enables to understand which method should we choose, if we have a limited number of genes, or when we want to reduce runtime and see how many genes are needed to achieve good results.

First, we computed the results of the baseline model - the random forest model without feature selection, and the other two classifiers without feature selection methods. As can be shown on Figures 4, 5 the feature selection process improves significantly the results. To receive the best parameters (feature selection method and number of genes) for each classifier, we summarized the best results obtained in cross-validation, by different evaluation metrics, as shown in Figure 3c. Since our goal is to detect liver disease, missing a sick patient (false negative) is more critical than misclassifying a healthy one (false positive), as it could delay diagnosis and treatment. In contrast, a false positive mainly leads to additional testing. Therefore, we prioritized maximizing the True Positive Rate (TPR) and evaluated our final results by taking the parameters which achieved the best TPR score(Figure 3c). The evaluation metric we chose to evaluate the classifier is a confusion matrix. As can be seen in figure 5, both the KNN and Mean Ensemble classifier were able to produce great results, both in TPR and TNR.

Our second task,Differentiating between different liver diseases, aimed to explore whether

there are distinguishable differences in the cfDNA profiles associated with different pathological conditions. Given our ability to identify whether a sample comes from a healthy or sick individual, it is crucial not only to recognize the illness but also to pinpoint the specific disease. This is the core goal of liquid biopsies, as they can potentially reduce the need for traditional biopsies, which are often invasive and carry associated risks. In contrast, liquid biopsies, typically based on a simple blood test, are non-invasive and carry a lower risk.

As shown in Figure 6b, our results suggest that there is indeed a possibility to differentiate between diseases using the cell-free ChIP data. This is particularly interesting because it implies that different pathologies cause distinct patterns of cell death, even within the same organ, such as the liver. However, we must also address the fact that misclassifications occurred. For certain cases, the way a disease alters the cfDNA composition might lead to patterns that overlap, with a different pathology, making differentiation more challenging.

Additionally, it is important to consider that the relatively small sample size used in this study may have impacted the results for the disease differentiation. As previously mentioned, we down sampled our data to ensure an equal number of occurrences for each diagnosis before splitting into training and testing sets. This down sampling was necessary due to the imbalanced number of samples across different diagnoses. But may have introduced a challenge for the classifiers. samples with valuable information that might have strengthened the classifiers were possibly removed. It is also possible that the method used for gene selection could be refined, as the adjusted p-value was not found to be significant. For future research we would like to find genes that their different expression is statistically significant. Furthermore, since the labeling was obtained through pathological examination of biopsies, we cannot be certain that all biopsies results are perfectly accurate, a problem which we didn't address.

In general, our findings show that plasma cfDNA can provide valuable information to identify liver disease, and even differentiate between different pathologies. However, it is essential to acknowledge some limitations that we believe this diagnostic approach has. One of the key challenges is that the composition of plasma cfDNA is influenced by cell death events since cfDNA is generally a byproduct of cell death. If a certain pathology does not involve an increased rate of cell death, we might not be able to detect it.

This raises an important biological question: Is it sufficient to rely solely on plasma cfDNA when attempting to detect diseases using liquid biopsies, or should we incorporate other signals derived from plasma, such as cfRNA, to improve detection?

Work Partition

Name	Tasks
Hillel Darvish	Preprocessing, Differentiation between different liver diseases
Ilay Gussarsky	Feature selection, Healthy vs. Sick classification
Amit Alper	Feature selection, Healthy vs. Sick classification, Cross Validation
David Gershelis	Feature selection, Biological conclusions
Ghasan Jaber	Healthy vs. Sick classification, Biological background

References

Sorbini, M., Carradori, T., Togliatto, G. M., Vaisitti, T., & Deaglio, S. (2024). Technical advances in circulating cell-free dna detection and analysis for personalized medicine in patients' care. *Biomolecules*, *14*(4), 498. https://doi.org/10.3390/biom14040498

Barrera-Saldaña, H. A., Fernández-Garza, L. E., & Barrera-Barrera, S. A. (2021). Liquid biopsy in chronic liver disease. *Annals of Hepatology*, *20*, 100197. https://doi.org/https://doi.org/10.1016/j.aohep.2020.03.008

Corcoran, R. B., & Chabner, B. A. (2018). Application of cell-free dna analysis to cancer treatment. *New England Journal of Medicine*, *379*(18), 1754–1765. https://doi.org/10.1056/NEJMra1706174

Jenuwein, T., & Allis, C. D. (2001). Translating the histone code. *Science*, *293*(5532), 1074–1080. https://doi.org/10.1126/science.1063127

Bannister, A. J., & Kouzarides, T. (2011). Regulation of chromatin by histone modifications. *Cell research*, *21*(3), 381–395. https://doi.org/10.1038/cr.2011.22

Sadeh, R., Sharkia, I., Fialkoff, G., Rahat, A., Gutin, J., Chappleboim, A., Nitzan, M., Fox-Fisher, I., Neiman, D., Meler, G., Kamari, Z., Yaish, D., Peretz, T., Hubert, A., Cohen, J. E., Salah, A., Temper, M., Grinshpun, A., Maoz, M., … Friedman, N. (2021). Chip-seq of plasma cell-free nucleosomes identifies gene expression programs of the cells of origin. *Nature Biotechnology*, *39*(5), 586–598. https://doi.org/10.1038/s41587-020-00775-6

Stonehouse, J. M., & Forrester, G. J. (1998). Robustness of the t and u tests under combined assumption violations. *Journal of Applied Statistics*, *25*(1), 63–74. https://doi.org/10.1080/02664769823304